

DEVELOPMENT OF A ROAD ACCIDENT SIMULATION FOR AUTOMATIC ROAD ACCIDENT REPORT SYSTEM

Keyword: Road Simulation, Accident Analysis, Human–Environment Interaction, Road Safety, Systems Theory

Abstract

Road accidents have become a crucial problem in Labo, Camarines Norte. Many road accidents happen because of lack of street lighting or the poor lighting, slippery roads, blind curves, and careless, sleepy, and tired drivers due to the longest highway that Labo has. This simulation study aims to develop a Road Accident Simulation and Automation Report System that can help predict and analyze accidents before it will happen. The Simulation Model uses math and physics formulas to recreate how accidents occur based on real factors that cause road accidents like vehicle speed, road condition, and angle of impact.

This study is guided by System Theory, which says that accidents happen because of the interaction between drivers, vehicles, roads, and the environment and not just one single cause. By studying these relationships, the simulation can help identify which are the dangerous areas around the place and test safety improvements, such as adding street lights or installing warning signs. As a result, this system will help local authorities in Labo, Camarines Norte make better plans to improve road safety and prevent future accidents.

Introduction

Road accidents are becoming a serious problem in Labo, Camarines Norte. Many accidents happen because of dangerous intersections, blind curves, and long highways. Some drivers also meet accidents for the reason that they are not familiar with the road, while others lose control due to slippery roads or dark areas with little or no street light. While in some cases, accidents occur when drivers become sleepy or tired, experience mechanical problems, and a lot of accidents happen because of the drivers under the influence of alcohol. Currently, there are no local tools to predict where and how accidents might occur, making it difficult for local authorities to prevent them. This study aims to develop a Road Accident Simulation and Modeling System for Automatic Road Accident Report System that can illustrate the possible accident scenarios. By using this system, we can understand the effects of vehicle speed and traffic lights on road safety and suggest strategies to reduce accidents.

Previous research has shown that simulation plays an important role in improving road safety. [1] published study developed a 3D road accident simulator that automatically generates accident reports and predicts driver responsibilities using machine learning. As a result, the system showed that simulations can produce realistic accident scenarios, generate training data for image recognition, and support authorities in managing accident information.

In another related study [2] published. It is about Statistical Mechanics and its Applications, the researchers used cellular automation to model and simulate car accidents at signalized intersections. It showed how traffic patterns and accidents can be analyzed with the use of simulation, which can provide predictions that will improve for traffic safety. This is relevant to the current study because it shows that even complex traffic interactions can be modeled to anticipate accidents, which is the principle applied in developing a local system for Labo, Camarines Norte.

By developing a simulation system and automatic road accident report, the local government and traffic authorities in Labo, Camarines Norte can create better plan safety strategies, such as improving street lighting, placing warning signs, and controlling vehicle speed in areas that are more likely to have accidents. This study aims to help residents and commuters stay safe by increasing awareness and improving road safety. While accident-prone areas can be closely monitored and managed. The system will serve as a useful tool to predict and prevent accidents, specifically the road safety problems in Labo, Camarines Norte.

Literature Review

Road safety has always been a major concern in many local areas Camarines Norte, where accidents are often caused by dangerous intersections, blind curves, slippery and dark roads, and careless and sometimes sleepy drivers due to long highways. To reduce the risk of road accidents, it is important to identify the locations that are more likely to have accidents and also it's very important to understand the factors that cause them. Recent studies have shown that simulation and modeling can be very useful tools in predicting and preventing road accidents. Researchers and engineers study different ways to identify dangerous areas and understand the causes of accidents to prevent road accidents. These research works provide valuable ideas that can help the future researcher and applied in developing the Road Accident Simulation and Modeling, which aims to help authorities monitor, analyze, and reduce accidents in the area.

The System Theory, introduced by Ludwig von Bertalanffy in 1968 explains that everything works as part of the whole system is made up of connected parts that can affect one another. When we apply it to road safety, this theory gives an idea that suggests accidents don't happen because of a single cause. Instead, they result from the interaction of several factors, such as the driver, the vehicle, the road condition, the weather, and even the traffic laws. Each of these components make up what is known as the Road Transport System (RTS), where the changes in one element can affect the others. In this development study, System Theory serves as the foundation for understanding how road accidents occur in Labo, Camarines Norte. By using simulation and modeling, it aims to analyze how different factors work together to create high-risk situations. This approach reflects the idea that improving road safety requires looking at the system as a whole rather than focusing on just one part. Within any system, the interconnectedness ensures that even the smallest change in one element can influence everything.

Simulations can demonstrate different factors working together to increase the risk of accidents. By identifying these patterns, local authorities can plan and design more effective interventions, such as adding more street lights, placing warning signs in blind curves, or enforcing speed limits to help prevent accidents and improve overall road safety. In the study of Hu (2024) which entitled *Estimating Accident-Prone Freeway Sections: Simulation and Accident Prediction Model*, they worked on identifying the most dangerous parts of a 77-kilometer freeway in Zhejiang Province, in China. They collected data about previous accidents and the physical structure of the road, such as its curves and the slopes. Two main statistical methods are shown in their studies, the improved cumulative frequency method and the accident matrix method were used to find which sections had the most accidents.

They also developed a new method that combined vehicle dynamic simulation using the CarSim software and speed consistency theory to estimate and understand accident-prone areas more accurately. Through this approach, they developed an accident prediction model that allows engineers to better understand how the road design and driver behavior can influence safety. The model can also be used to assess the safety of new road layouts before they are constructed. This is connected to the current study, as the proposed system in Labo also aims to use simulation to identify high-risk areas and support improvement of local road design and safety plans.

In a similar study, Cheng (2022) examined vehicle- pedestrian accidents using simulation and measured the risk of injury through the Head Injury Criterion (HIC). The results showed that factors such as vehicle speed, collision angle, and vehicle can greatly affect how severe an accident can be. This research supports the idea that simulation can be a useful tool for identifying risk factors and testing safety measures.

In another related study, Kizito (2020) explored the cause of road accidents through their research entitled *Modeling the Complexity of Road Accidents Prevention: A System Dynamics Approach*. They viewed accidents as a part of a larger system known as the Road Transport System (RTS). Instead of blaming only the drivers, their findings showed that accidents are caused by many connected factors such as road conditions, vehicle maintenance, traffic management, and human behavior which is also connected to the System Theory. Their study used Dynamic Synthesis Methodology (DSM) to analyze these interactions and develop models showing how accidents happen before a crash occurs. Data from police, road users, and engineers helped them identify the main areas where safety interventions could reduce accidents. This approach relates to the Simulation and Modeling of a Road Accident in Camarines Norte because it also recognizes that accidents are not just caused by a single factor. By considering human error, road design, lighting, and environmental conditions, the simulation and modeling to be developed will adopt a multi-factor approach similar to Kizito and Semwanga's model to analyze local conditions and predict where accidents are likely to occur.

Additionally, in the paper published “AI and IoT-Based Road Accident Detection and Reporting System” by Patil et al., (2023), the study introduces a system that combines vibration sensors, accelerometers, alcohol sensors, GPS, camera, and Raspberry Pi to detect collision. Once there’s an accident, the system will automatically send data to the cloud for reporting and enable a faster emergency response. Similarly, Akshatha (2017), in their study “Automatic Accident Detection and Intelligence Navigation System” proposed a system that not only detects accidents but also guides an ambulance to the nearest hospital using an intelligent navigation feature. The system will help prevent delays caused by traffic congestion by transmitting the accident’s GPS location and the victim’s information to hospitals and ambulances through the use of e-NOTIFY system, using vehicle-to-vehicle communication to clear routes for emergency transport.

Moreover, combining the machine learning techniques with three-dimensional simulation environments has become a powerful way to predict and analyse road accidents. Yawovi (2020) developed a system based on advanced Random Forest algorithms that could automatically generate detailed accident reports and accurately predicted driver responsibilities. Their simulator generated realistic and complex accident scenarios that can give a lot of purpose, including training image recognition models and providing valuable information for accident management teams. With the use of this system it made accident data more accurate and complete, it can also reduce the need of manual analysis. Overall, the study shows how the use of data-driven machine learning and advanced simulation can fill important gaps in accident prediction and offer practical tools that can make roads safer.

All of these studies have the same goal. Which is to make the roads safer through the use of advanced technologies such as simulation, data modeling, automation, and artificial intelligence. With the help of these previous studies, analyzing data and simulating or understanding the different scenarios, researchers can better understand how and what are the possible reasons why accidents happen and what can be done to prevent them. For Labo, Camarines Norte, where many accidents happen due to slippery roads, blind curves, and poor lighting, adopting these approaches will be very important toward developing a Road Accident Simulation and Modeling System. This system will not only help predict where accidents are likely to occur but also test the effectiveness of safety measures such as improved street lighting, better signage, and stricter law enforcement ultimately helping create safer roads and protect the lives of local residents.

Conceptual Framework

This section illustrates how human, vehicle, environmental, and road factors work together as part of the system that leads to accidents. Based on System Theory, it emphasizes that accidents arise from the interaction among these interconnected elements rather than from a single, isolated cause.

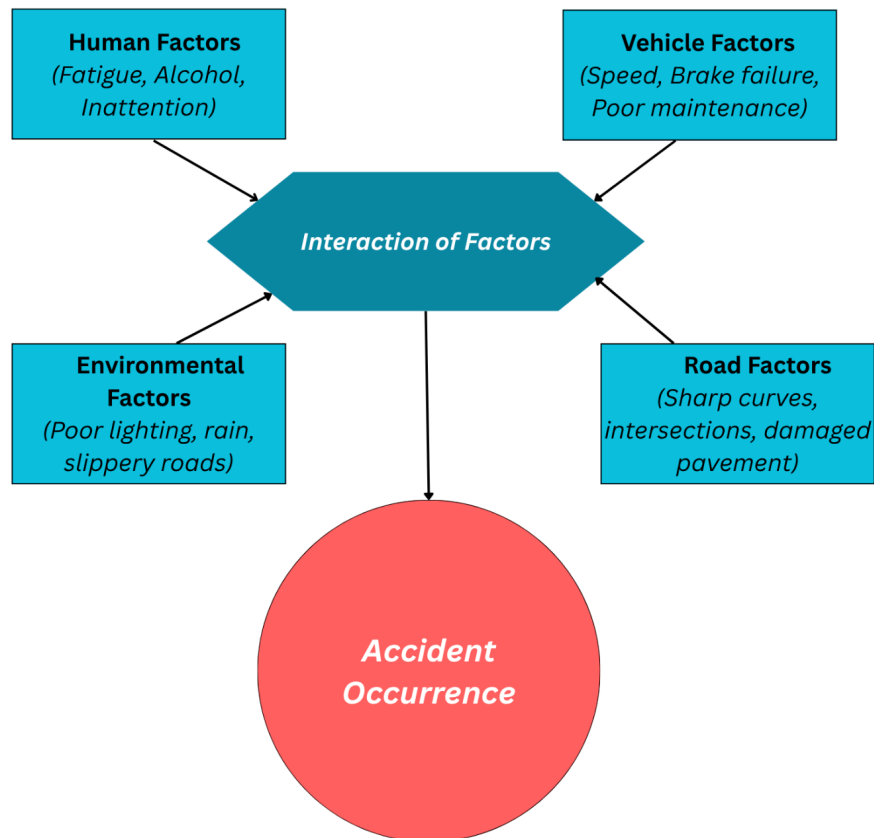


Figure 1. Conceptual Framework (Based on System Theory)

MATHEMATICAL MODEL PROCESS

This development uses two linked models to turn real-world accident factors into clear, testable results. The first Model is the Simulation; it recreates the physical events of a crash using physics and motion equations. The second model is the Reporting; it will turn the simulation's raw numbers into a readable, structured accident report. Below is a brief, human-friendly mathematical description of how those models work together and the main formulas they use.

Overview of the Process

1. **Collect Input Data** – Start by gathering important details such as the speed of each vehicle, their weight, the angle of impact, the condition of the road (like

how slippery it is), lighting, and the drivers' reactions before the crash.

2. **Run the Simulation Model (Model 1)** – Using the collected data, the system calculates how each vehicle moved before the crash, what happened during the collision, and how they behaved afterward. It applies mathematical principles like momentum and energy to make the simulation realistic.
3. **Generate Raw Results** – After the simulation, the system produces data such as the final positions of the vehicles, their speeds after the collision, the force of impact, their movement paths, skid marks, and the timing of each event.
4. **Run the Reporting Model (Model 2)** – The raw data is then analyzed and converted into a clear, human-readable report. This model identifies what type of accident occurred (for example, a rear-end collision) and rates its severity based on the simulation results.
5. **Validate and Refine** – Finally, the simulated results are compared with real-life accident data, if available. Adjustments are made to improve the model's accuracy and reliability for future simulations.

Model 1: The Simulation Model

The use of this model is to simulate what happens during a road accident with the use of mathematical formulas. It uses input data such as vehicle speed, weight, angle of impact, and road conditions. The output of this model includes numerical results like the vehicle's final position, the force of impact, and the direction the vehicles moved after the crash.

Mathematical formulas such as:

- **Displacement:** $d = v_i t + \frac{1}{2} a t^2$
- **Final Velocity:** $v_f = v_i + a t$
- **Momentum Conservation:** $m_1 v_{1i} + m_2 v_{2i} = (m_1 + m_2) v_f$

These will help calculate how vehicles move, brake, and collide. The principle of conservation of momentum is especially important because it shows that the total momentum before and after the crash remains the same, even though energy is lost during impact.

Model 2: The Reporting Model

Once the simulation produces the raw data, the reporting model processes this information to create a clear, human-readable report. It automatically writes a

summary of the accident, includes diagrams, and classifies the type and severity of the collision.

This model uses decision rules and state logic to describe what happened:

- For example:
 - *If the front of one car hits the rear of another, then it's a "rear-end collision."*
 - *If the change in speed is very high, then the crash severity is "high."*

By using and combining these models together, the system can both simulate and explain an accident accurately. This makes the process faster, reduces human error, and helps authorities analyze the causes of road accidents more effectively.

System Architecture

The system architecture shows how the proposed simulation and reporting model works from data input to final output. It illustrates how information is processed through simulation, data analysis, and reporting stages. The structure highlights the integration of simulation and reporting for reliable analysis.

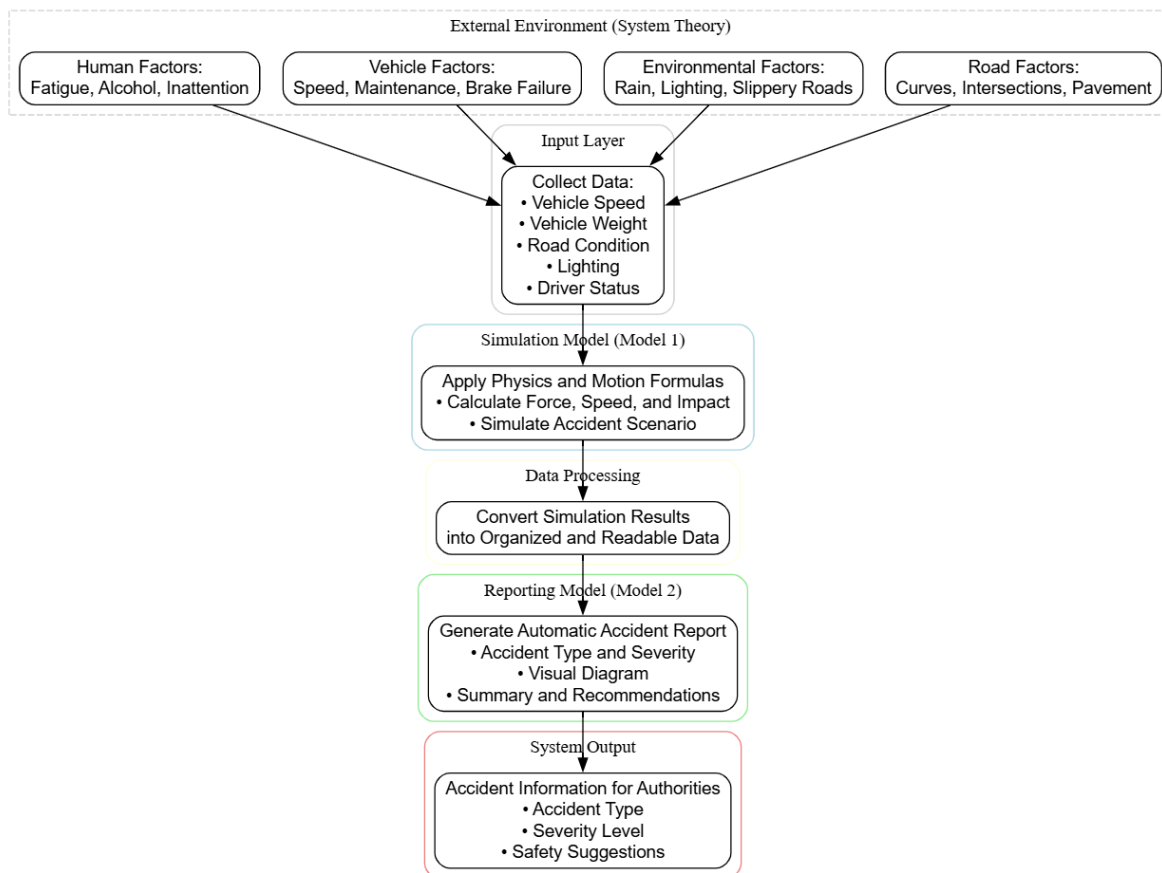


Figure 2. System Architecture of the Proposed Simulation and Reporting Model

RESULTS AND DISCUSSION

This section presents the outcomes generated by the Road Accident Simulation and Automatic Reporting System, highlighting various factors such as Human Factors, Vehicle Factors, Environmental Factors, and Road Factors can affect accident severity. The simulation processed different sample scenarios representing common roadway situations around Labo, Camarines Norte. The system automatically computed hazard indicators, generated risk levels, and generated summary reports to support early response in decision making.

Overall, the findings show that when the road conditions become challenging such as poor lighting, bad weather, or unsafe road conditions and even when the drivers speed up or lose focus, the likelihood of serious accidents increases significantly. The results also highlight how these factors work together as a part of a system rather than independently reflecting the core idea of System Theory. In this view, accidents do not happen in just one mistake or condition; instead, they come from the combined interaction of several subsystems, including the driver, the environment, the vehicle and the road itself.

1. Sample Simulation Scenarios

Table 1. Sample Scenarios and Computed Severity Scores for Labo, Camarines Norte

This table shows selected scenarios and their severity scores. High severity scores were seen in scenarios involving high speeds, fatigue and poor road conditions (S3, S5, S8). However, low severity scores occurred when drivers stayed calm and road conditions were manageable (S9), even in the presence of minor obstructions. Scenarios such as S1, S4, S7, and S10 demonstrate that even at low speeds, human and road elements can combine to produce high-risk outcomes.

Scenario	Road Condition	Lighting	Speed (km/h)	Driver Behavior	Obstacle	Severity Score	Risk Level
S1	Wet	Low light	60	Distracted	None	75	High
S2	Dry	Daylight	45	Alert	Pedestrian	50	Moderate
S3	Flooded	Low light	80	Fatigued	Debris	92	Critical
S4	Muddy	Night	70	Drowsy	Animal	85	High

S5	Dry	Night	100	Reckless	Vehicle	95	Critical
S6	Wet	Daylight	55	Alert	None	60	Moderate
S7	Rough Road	Low light	65	Distracted	None	80	High
S8	Dry	Daylight	120	Reckless	None	98	Critical
S9	Flooded	Daylight	35	Calm	Debris	45	Low
S10	Muddy	Night	75	Fatigued	None	87	High

2. Distribution of Severity Scores

Table 2. Severity Category Distribution in Simulated Scenarios

The distribution of severity scores across all simulated scenarios is shown in table 2. The majority of scenarios (60%) were rated as high or critical, highlighting the fact that accidents are frequently caused by multiple interacting factors. Only 20% of scenarios were classified as low risk, highlighting the need for proactive safety measures and interventions.

Severity Category	Score Range	Count	Percentage
Low	0–49	1	10%
Moderate	50–70	2	20%
High	71–90	4	40%
Critical	91–100	3	30%

Note: Majority of scenarios (60%) fall under High or Critical severity, emphasizing the need for proactive measures.

3. Impact of Individual Factors

Table 3. Average Severity by Driver Behavior

The severity of the crash was substantially impacted by driver behavior. Fatigue and irresponsible driving were linked to the highest severity scores, but alert and calm driving considerably decreased risk. This shows the importance of human factors in accident prevention.

Driver Behavior	Avg. Severity	Interpretation
Alert	55	Safe driving reduces risk even on moderately challenging roads.
Distracted	77	Distraction significantly increases severity, especially at night or on wet roads.
Fatigued	89	Fatigue is among the highest contributors to accident risk.
Reckless	97	Reckless driving consistently results in critical accidents.
Calm	45	Minimal risk occurs even with minor obstacles.

Table 4. Average Severity by Road Condition

Road conditions have a substantial impact on accident outcomes. Muddy and rocky roads presented the greatest risk due to low traction, while dry roads posed a moderate risk, depending on driver behaviour and speed. Flooded roadways enhanced risk, especially when combined with high speeds or driver tiredness.

Road Condition	Mean Severity
Dry	81
Wet	68
Flooded	69
Muddy	86
Rough Road	80

4. Visualization of Factor Interaction

The simulation showed that accident severity increased with speed. Poor lighting, poor road conditions, and driver exhaustion all increase risk, suggesting that high-severity scenarios are the consequence of a complex interaction of numerous factors rather than a single cause.



5. System Reporting and Decision Support

Figure 2. Sample Automatic Accident Report

The technology automatically generates structured accident reports based on driver behavior, vehicle speed, road conditions, and recognized impediments. These reports make actionable recommendations to authorities, resulting in faster emergency response and better traffic management.

6. System Theory Perspective

The simulation findings proved the interconnection of the various subsystems. Human variables such as driver fatigue, distraction, and recklessness, along with vehicle speed, environmental conditions, and road surfaces, resulted in the highest severity ratings. High-risk scenarios occurred predominantly when two or more subsystems failed at the

same time, supporting the System Theory tenet that accidents are emergent occurrences caused by interacting elements rather than isolated faults.

7. Implications for Road Safety in Labo, Camarines Norte

The findings indicate that high-risk roads, particularly those that are muddy, rough, or flood-prone, require sufficient maintenance, enough lighting, and prominent warning signs. Speed-monitoring and driver fatigue awareness initiatives are crucial. The simulation results can help local government entities prioritize resources for accident prevention and speedy response. Automated reporting enables timely intervention, reducing casualties and preventing secondary incidents.

Conclusion

The Road Accident Simulation and Automatic Reporting System successfully demonstrated how human, vehicle, environmental, and road elements interact to determine accident likelihood and severity in Labo, Camarines Norte. Using physics-based motion and momentum calculations, the system was able to simulate realistic accident scenarios, quantify risk levels, and produce clear, structured reports. The findings indicated that high-risk situations are most likely to occur when numerous factors fail at the same time, validating System Theory's notion that accidents are emergent outcomes of interrelated elements rather than isolated events.

The study also showed that human variables, particularly driver fatigue, distraction, and reckless behavior, had a considerable impact on accident severity, while road conditions such as mud, rough surfaces, and poor lighting increase risks. By recognizing these high-risk scenarios, the technology provides local governments with actionable insights for implementing targeted safety interventions such as better street illumination, warning signage, speed restriction, and driver awareness programs. Overall, the system provides a useful data-driven tool for accident prevention, emergency response, and road safety planning in Labo, Camarines Norte.

Future Works

- 1. Real-Time Data Integration - Use actual traffic, weather, and vehicle sensor data to improve simulation accuracy and allow for real-time accident prediction.**
- 2. Expansion to AI-Based Analysis: Use machine learning algorithms to automatically classify accident types, anticipate likely causes, and offer preventive measures.**
- 3. 3D Visualization of Accidents: Implement 3D animations of simulated accidents to enable authorities better visualize scenarios for training and planning purposes.**

4. **Mobile and web-based accessibility** - Create mobile or online interfaces that enable field officers and local governments to access simulation results and reports remotely.
5. **Integration with Emergency Response Systems** - Connect the system to local emergency units to automatically inform responders and hospitals, resulting in speedier assistance.
6. **Enhanced Multi-Factor Modeling** - To improve prediction accuracy, include variables including vehicle repair status, driver health, and traffic congestion patterns.
7. **Community Awareness Features** - Create public dashboards to highlight accident-prone locations and encourage citizens to drive more safely.

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